

How Much Adaptation Is Enough?

Drift Geometry and Minimal Calibration in Brain-to-Text BCI

Anonymous Authors

Abstract

Brain-to-text BCI decoders must remain usable as neural signals change across recording sessions. We audit how much input adaptation is needed when the official T15 brain-to-text GRU is evaluated across sessions. Native day-specific input layers reproduce low validation phoneme error (9.06% PER), while forcing sessions through non-native input layers increases PER to 22.67–34.43%. Unsupervised moment corrections fail to recover this loss. A supervised input-layer adapter trained from 20 labeled calibration trials improves the fixed-middle stress setting from 22.67% to 18.59% PER, but still leaves substantial degradation. The strongest practical result is source-layer selection: using a short unlabeled beginning-of-day window to choose a past input layer reaches 12.74% PER at $K = 20$, while the immediately previous session is even stronger at 12.21%. Secondary geometry-only analyses of T12 diagnostic blocks and inner-speech behavior blocks show similar state structure beyond the main T15 decoder experiment. These results suggest that recency should be the default adaptation prior, while drift geometry is a useful signal for deciding when that default may need to be overridden.

1 Introduction

High-performance speech BCIs commonly include session-specific adaptation machinery because intracortical neural features drift across days. Long-term stability has long been recognized as a central challenge for practical BCI deployment [Vaadia and Birbaumer, 2009]. In this work, “drift” means that the neural feature distribution recorded on a new day no longer matches the distribution seen by the decoder during earlier sessions. A practical BCI therefore needs a way to decide whether yesterday’s input mapping is still usable, whether an older mapping is a better match, or whether a short calibration step is needed. This raises a practical question:

before retraining a decoder or collecting a long supervised calibration set, how much adaptation is enough? Our main practical finding is that cross-session BCI drift is often better handled by choosing the right input state than by retraining the decoder.

We examine this question in the official T15 brain-to-text GRU baseline from a rapidly calibrating speech neuroprosthesis study [Card et al., 2024]. The model contains learned day-specific input layers, followed by a GRU and output head. Intuitively, each input layer acts like a small session-specific adapter that maps that day’s neural features into the representation expected by the shared GRU decoder. This structure creates a useful stress test: if a target session is decoded through the wrong input layer, performance measures the decoder-facing cost of cross-session input mismatch. We then ask whether simple feature corrections, small learned adapters, or geometry-guided source selection can recover the lost performance.

The contribution is not a new speech decoder. Instead, we provide a decoder-facing adaptation audit:

- Wrong input states break the decoder: native-day PER is 9.06%, while fixed non-native input layers increase PER to 22.67–34.43%.
- Simple mean/scale correction is insufficient; input-layer calibration with $K = 20$ improves PER to 18.59% but remains far from native-day.
- Previous-session reuse is the strongest simple policy at $K = 20$ (12.21% PER), while geometry selection is close (12.74%) and reveals older-state candidates.
- Geometry is a retrieval signal, not a standalone policy: at $K = 20$ it selects older states in 11 sessions, but wins only 3 times, motivating a confidence gate.

We do not claim that neural drift geometry is new. Rather, we show that in a deployed-style brain-to-text decoder with learned day-specific input layers, short-window geometry is actionable for choosing among existing input-layer states and for framing recency as an explicit adaptation prior. Figure 1 summarizes this adaptation-ladder view.

Recency-aware adaptation ladder for cross-session BCI drift

Estimate new-session geometry → retrieve a compatible input state → decode with the GRU frozen

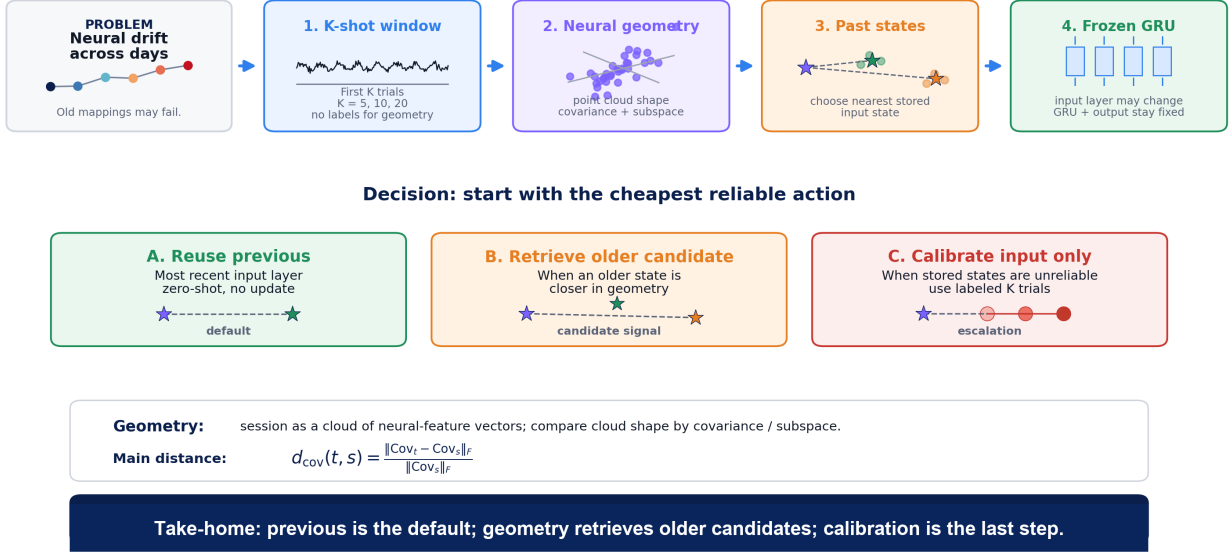


Figure 1: Recency-aware adaptation ladder for cross-session BCI drift. At the start of a new target session, the first K trials are used to estimate neural feature geometry. The target geometry is compared to a library of past input states. The system first reuses the previous input layer as the default, retrieves an older candidate when geometry suggests a closer match, or performs lightweight input-layer calibration when stored states are unreliable. The shared GRU decoder and output head remain frozen throughout.

2 Related Work

Intracortical communication BCIs have advanced from handwriting-based brain-to-text decoding [Willett et al., 2021; Fan et al., 2023] to speech neuroprostheses that decode attempted speech [Willett et al., 2023; Card et al., 2024]. Recent work also explores stronger speech backbones and pretrained acoustic representations for neural decoding [Fiedler et al., 2025]. Our work is complementary: given an already trained brain-to-text decoder with day-specific input layers, we ask how to keep the decoder usable across sessions through lightweight input-side adaptation rather than retraining the backbone.

Cross-session stability is often studied through neural geometry and alignment. Manifold-alignment methods can stabilize fixed decoders across recording instabilities [Degenhart et al., 2020], and dynamics-aware approaches such as NoMAD use temporal structure for unsupervised alignment over weeks to months [Karpowicz et al., 2025]. Speech BCI work also explores adversarial session-invariant learning [Zhang et al., 2026] and continual recalibration with pseudo-labels [Fan et al., 2023]. We do not propose a new invariant decoder or full recalibration method. Instead, we audit low-cost choices available before retraining: moment correction, lightweight input calibration, previous-session reuse, and geometry-guided source-layer retrieval.

3 Data and Decoder

T15. The main experiments use T15 copy-task neural features from the official brain-to-text dataset [Card et al., 2024]. We evaluate 41 validation sessions with ground-truth phoneme labels. Neural inputs are passed through the official GRU checkpoint and decoded greedily at the phoneme level. During evaluation, repeated symbols and blank tokens are removed using the standard CTC collapse rule before computing PER. This collapse step is only part of decoding/evaluation; learned calibration is trained separately using CTC loss.

Official baseline. The decoder contains learned day-specific input layers, a GRU backbone, and an output layer over phoneme classes. In the native-day condition, each session uses its own input layer. In cross-day stress tests, target sessions use a fixed or selected non-native source input layer while the rest of the decoder remains unchanged.

Evaluation splits. The main T15 decoder-facing experiments use the validation/test sessions provided with the official dataset. In this split, sessions contain a limited number of held-out trials, so our beginning-of-day experiments treat $K \leq 20$ as the main minimal-calibration regime: the first K trials are used for geometry estimation or supervised input calibration, and all reported PER is computed only on the remaining trials. Larger windows, such as $K = 40$, are useful as exploratory feasibility checks but consume a large frac-

tion of this split and are therefore not used as the main claim. T12 and inner-speech experiments are used only as geometry-only supporting analyses and are not evaluated with decoder PER.

T12 diagnostic blocks. As a secondary check, we analyze diagnostic blocks from the T12 high-performance speech neuroprosthesis dataset [Willett et al., 2023]. This is a geometry-only analysis of spike-power and threshold-crossing feature variants; we do not reproduce the T12 decoder or report T12 PER.

4 Methods

4.1 Decoder-Facing Metrics

For each T15 validation trial, we compute greedy phoneme predictions from the CTC output stream and then apply CTC collapse before computing phoneme error rate (PER). We report phoneme-weighted PER unless otherwise stated. Quality diagnostics such as blank rate, confidence, and entropy are saved, but the core results use PER.

For a set of trials $\mathcal{D}$, the reported PER is

$$\text{PER}(\mathcal{D}) = \frac{\sum_{i \in \mathcal{D}} \text{edit}(y_i, \hat{y}_i)}{\sum_{i \in \mathcal{D}} |y_i|}.$$

We avoid mixing PER with derived recovery percentages in the main result tables: lower PER is always better, and improvements are reported as direct PER changes. This keeps the comparison clear when learned-adaptor experiments evaluate on remaining validation trials after holding out the first K calibration trials.

4.2 Cross-Day Stress Test

We compare native-day decoding against three fixed non-native source layers: early (t15.2023.08.13), middle (t15.2023.11.26), and late (t15.2025.04.13). This tests whether the decoder degradation is tied to one bad source layer or is a general consequence of input-layer mismatch.

4.3 Adaptation Ladder

Using the middle source layer, we test no correction, target z-scoring, source moment matching, diagonal affine calibration, and full input-layer calibration. Moment methods use full-session unsupervised feature statistics. Learned adaptors use the first K labeled validation trials and are evaluated only on the remaining trials. We use K to simulate a beginning-of-day calibration window: the first K trials are used for adaptation, and the rest of the session is held out for evaluation.

The two unsupervised moment corrections are deliberately simple. Target z-scoring maps x to $(x - \mu_t)/(\sigma_t + \epsilon)$, while source moment matching maps target features into the source marginal statistics:

$$x' = \frac{x - \mu_t}{\sigma_t + \epsilon}(\sigma_s + \epsilon) + \mu_s.$$

These baselines test whether the day-specific input layers can be approximated by channel-wise distribution alignment. Diagonal affine and input-layer calibration are supervised small-calibration probes: the GRU and output head remain frozen, and only a small input transform is learned from the first K validation trials using CTC loss.

4.4 Drift Geometry and Source Selection

Neural feature geometry. We represent each session as a cloud of neural feature vectors in channel space. Each time bin from each trial is one point in this space. The shape of this cloud is summarized using covariance and low-dimensional subspace statistics. For K -shot source selection, the first K unlabeled target trials are used to estimate the target covariance. We then compare this covariance to each past source session using relative Frobenius covariance distance and reuse the input layer of the closest past session.

For each source-target pair, we compute temporal distance, mean and scale shifts, relative covariance shift, CORAL-style distance, and PCA subspace distances. We then select a source input layer by nearest covariance geometry. To approximate deployment, a K -shot variant estimates target geometry from only the first $K \in \{5, 10, 20\}$ unlabeled validation trials and evaluates on the remaining trials. We focus the main claims on $K \leq 20$ because this represents the minimal-calibration regime in the available validation split.

Because temporal recency is an obvious baseline, we also evaluate the immediately previous input layer. Finally, we test a simple recency-aware override: use the previous input layer by default and switch to the geometry-selected older source only if its covariance distance is less than α times the distance to the previous source.

The K -shot source-selection rule is unsupervised with respect to labels. Let $S_t^{(K)}$ denote statistics estimated from the first K target trials, and let S_s denote source-session statistics estimated from past data. The geometry-selected source is

$$s^*(t) = \arg \min_{s \in \mathcal{S}_{\text{past}}(t)} d_{\text{cov}}(S_t^{(K)}, S_s),$$

where d_{cov} is relative Frobenius covariance shift. We compare this to a previous-session policy, $s_{\text{prev}}(t)$, because recency is the strongest prior available in a longitudinal BCI.

4.5 Input-State Library Policies

The source-selection experiments can be interpreted as retrieving an input state from a memory of previous sessions. We therefore compare four policies: using only the previous session, selecting geometrically from the last 3 sessions, selecting from the last 5 sessions, and selecting from all past sessions. This ablation asks whether a practical system needs to retain all historical input layers, or whether a bounded recent library is enough under the current geometry metric.

Table 1: T15 decoder-facing stress test and adaptation ladder. Mean, best-session, and worst-session values are phoneme-weighted PER. Fixed-source and moment-correction rows use the middle source t15.2023.11.26. Learned adapter rows use $K = 20$ calibration trials and are evaluated on remaining trials. Lower PER is better.

Condition	Mean PER ↓	Best session	Worst session
Native-day	9.06%	0.80% (11-19)	23.82% (03-14)
Fixed early	28.81%	7.38% (08-13)	58.28% (03-14)
Fixed middle	22.67%	1.20% (11-19)	56.44% (03-30)
Fixed late	34.43%	16.40% (01-12)	49.04% (07-19)
Moment match	22.73%	1.20% (11-19)	56.90% (03-30)
Diag. affine, $K = 20$	20.21%	1.94% (11-26)	53.26% (03-30)
Input layer, $K = 20$	18.59%	2.46% (11-26)	52.23% (03-30)

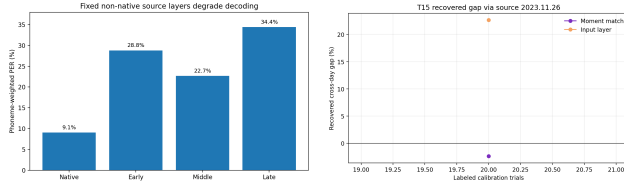


Figure 2: Left: fixed non-native source layers strongly degrade T15 phoneme decoding relative to native-day decoding. Right: small learned calibration improves over the fixed-middle source but remains far from native-day decoding.

5 Experiments

We organize the empirical study into five experiments, each paired with a table or figure. The T15 experiments are the main decoder-facing results; T12 and inner-speech analyses are supporting geometry-only checks. The experiments follow an adaptation ladder: fixed source → moment correction → learned calibration → source selection → recency/library policies → geometry-only supporting checks.

5.1 Cross-Day Stress Test and Adaptation Ladder

Native-day decoding reaches 9.06% phoneme-weighted PER, confirming that the local evaluation reproduces the expected behavior of the official checkpoint. When all sessions are forced through non-native source input layers, PER rises to 22.67–34.43%, harming 40/41 sessions for each source choice.

The natural first hypothesis is that cross-session mismatch is mostly a mean/variance shift. The data do not support this. Target z-scoring and moment matching to the source distribution remain essentially tied with no correction, and are slightly worse in aggregate. This is consistent with neurobiological evidence that long-term sensorimotor learning involves population-level reorganization rather than only independent single-channel gain changes [Mandelblat-Cerf et al., 2011]. Together, these results suggest that the learned day-specific input layers capture transformations richer than channel-wise mean and variance.

Supervised diagonal affine and input-layer calibration improve PER beyond moment matching. With $K = 20$,

Table 2: Beginning-of-day source selection on remaining validation trials. Previous-session recency is the strongest simple baseline, while K-shot geometry selection remains close and often selects non-previous sources. Lower PER is better.

Method	$K = 5$	$K = 10$	$K = 20$
Native-day	8.95%	8.88%	8.90%
Fixed middle	22.51%	22.40%	22.30%
Previous source	13.09%	12.84%	12.21%
K-shot geometry source	14.00%	13.19%	12.74%
Best simple override	13.09%	12.90%	12.18%

diagonal affine reduces weighted PER from 21.51% to 20.21%. A longer CUDA input-layer calibration run improves the learned input layer to 18.59%, compared with 22.67% for the fixed-middle source and 9.06% for native-day decoding. A follow-up initialization check suggests that calibration should be treated as residual adaptation rather than reset: on the same $K = 20$ subset, initializing from the previous input state improves 11.58% to 11.03%, whereas fixed-middle initialization remains much weaker at 17.93%. Thus learned input adaptation is a useful rung of the adaptation ladder, but even this stronger small-calibration setting does not replace a well-matched prior input state.

5.2 Beginning-of-Day Source Selection

Geometry-guided source selection performs much better than directly adapting a fixed middle source. In an offline all-candidate setting, choosing the nearest non-native input layer by covariance geometry reaches 11.38% PER. In a stricter past-only setting, it reaches 13.58% PER on 40 sessions, compared with 22.76% for the fixed middle source on the same sessions.

The beginning-of-day version is more realistic. Here, K denotes the first K beginning-of-day trials used for calibration or geometry estimation; evaluation is performed only on the remaining trials. With this protocol, past-only source selection reaches 14.00%, 13.19%, and 12.74% PER at $K = 5, 10, 20$, far below the fixed-middle PER of 22.51%, 22.40%, and 22.30% on the same remaining-trial subsets. However, the immediately previous input layer is even stronger: 13.09%, 12.84%, and 12.21% PER on the same subsets. This prior has a time horizon: previous-source PER is 9.97% when the last available input state is 0–3 days old, 13.75% at 4–14 days, and 24.54% beyond two weeks. Thus previous is a strong default only when it is truly recent. Figure 3 visualizes the pairwise geometry underlying this source-selection problem.

5.3 Reusable Input-State Libraries

The previous-session input layer is the strongest simple baseline, but it is not always the geometrically nearest source. In T15, K-shot geometry selection chooses the previous session in most targets, but selects an older non-previous input layer in 22–30% of cases. Depending on whether improvement is summarized by trial-mean PER

T15 longitudinal geometry: days close in time are usually close in input-state geometry

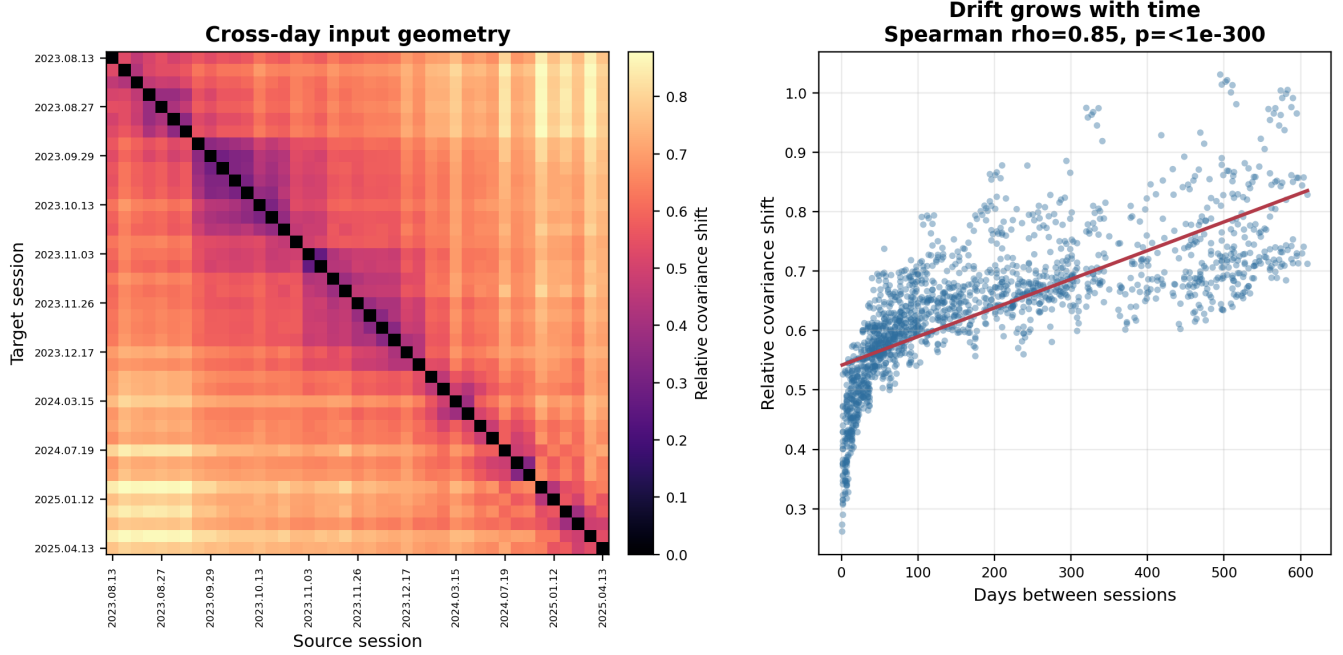


Figure 3: T15 longitudinal geometry map. Left: pairwise relative covariance shift between source and target sessions, ordered by date, shows a diagonal band of nearby sessions with similar input geometry. Right: covariance shift increases with temporal separation across source-target pairs (Spearman $\rho = 0.85$), indicating that session geometry drifts with time while retaining non-monotonic structure.

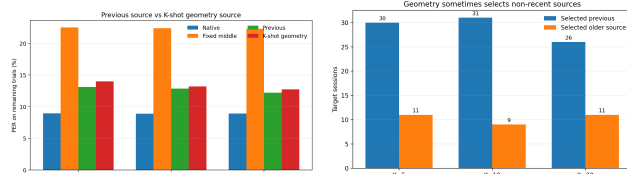


Figure 4: Left: previous-session recency is a strong source-selection baseline and slightly outperforms unconditional K-shot geometry selection. Right: geometry still reveals non-recent source matches, suggesting sessions are not always closest to the immediately previous day.

or phoneme-weighted session PER, these older sources outperform the previous-session source in a small minority of non-previous selections. This suggests that some target sessions may resemble older input-layer states rather than only drifting monotonically away from the previous day.

To inspect these non-previous selections directly at $K = 20$, we compare the geometry-selected source to the immediately previous source for each target session. Geometry selects the previous source in 26 sessions and an older non-previous source in 11 sessions. Among those 11 older selections, the older source achieves lower PER in 3 cases, the previous source remains better in 7 cases, and 1 case is tied. Thus geometry identifies plausible older-state candidates, but covariance distance alone is

not sufficient as a deployment policy.

One possible explanation is that a day-specific input layer may represent a neural recording state rather than a purely chronological day. Changes in recording quality, electrode yield, thresholding, neural strategy, or user state could move a new session closer to an older feature geometry than to the immediately preceding session. This view is also consistent with evidence that local cortical network activity can be modulated by action-perception state [Engelhard et al., 2019]. In this view, source selection is closer to memory retrieval over past input states than to choosing the most recent date.

The practical implication is that a longitudinal BCI should not keep only the latest input layer. A more robust system could maintain a small library of past input-layer states, use the previous session as the default, and invoke a recency-aware geometry gate only when the beginning-of-day neural geometry strongly suggests that an older state is a better match.

We therefore asked how large this past-state library needs to be. For each target session, we restricted K-shot geometry selection to the previous session, the last 3 sessions, the last 5 sessions, or all past sessions. The immediately previous session remains the best aggregate policy, but a small recent library is close to the all-past library. At $K = 20$, last-5 and all-past selection both reach 12.74% PER, while last-3 gives 12.76%. Expanding the nearest-source search to all past layers does not

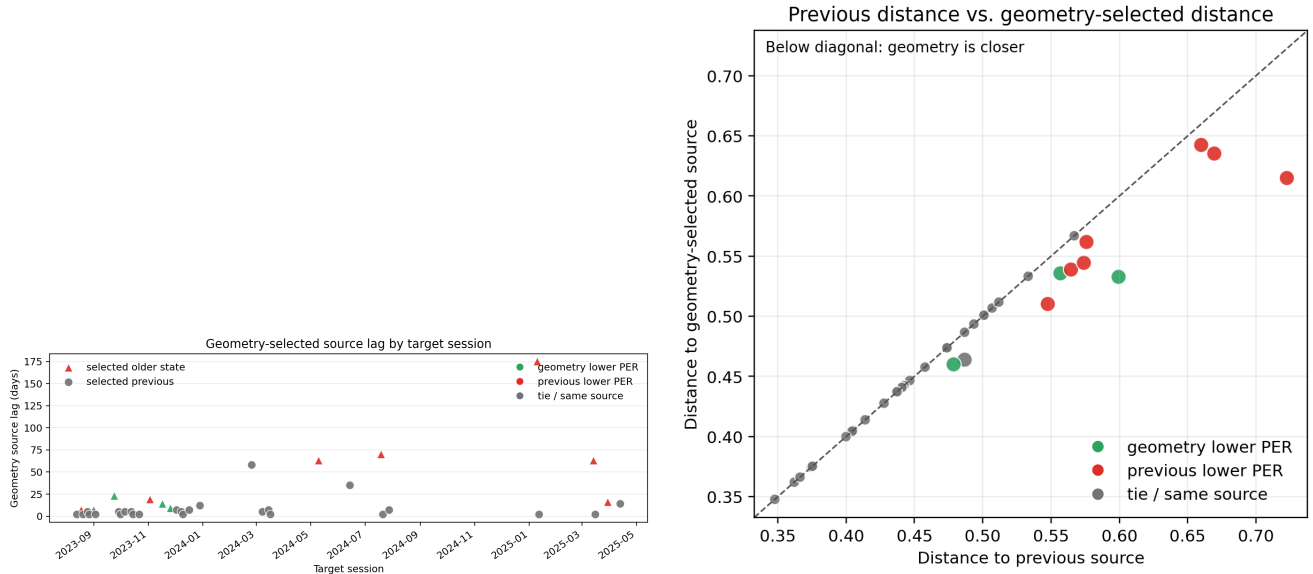


Figure 5: Geometry-selected older sources are meaningful candidates but not reliable standalone decisions at $K = 20$. Left: most target sessions select the immediately previous session, but several jump to older stored input states. Right: geometry-selected sources are closer in covariance distance by construction, but lower geometry distance does not always translate to lower PER. Green points indicate cases where the geometry-selected source outperforms previous; red points indicate cases where previous remains better.

Table 3: K-shot input-state library-size ablation. PER is measured on remaining validation trials after using the first K trials to estimate target geometry. A bounded recent library nearly matches all-past selection, while previous-only recency remains the strongest aggregate baseline.

K	Previous	Last 3	Last 5	All past
5	13.09%	13.47%	13.55%	14.00%
10	12.84%	13.19%	13.19%	13.19%
20	12.21%	12.76%	12.74%	12.74%

Table 4: Leave-one-session-out learned gate. The gate defaults to the previous source and learns a one-feature decision stump for when to override with the geometry-selected older source. It gives only a tiny gain at $K = 20$. Lower PER is better.

K	Previous	Geometry	Gate	Overrides
5	13.09%	14.00%	13.37%	6
10	12.84%	13.19%	13.18%	5
20	12.21%	12.74%	12.15%	4

improve aggregate PER over recency under this simple distance rule; rather, it increases the need for a confidence gate before using older states.

5.4 Gating and Drift Fingerprints

This experiment asks why geometry is useful but hard to deploy: can it safely override recency, and when do older states actually help? Since geometry sometimes chooses older sources, we tested whether a simple distance-ratio gate can improve over recency: keep the previous source unless the geometry-selected source is at least an α fraction closer in covariance distance. This rule does not yet produce a meaningful improvement. At $\alpha = 0.9$, it ties previous at $K = 5$, is slightly worse at $K = 10$ (12.90% vs 12.84%), and improves only marginally at $K = 20$ (12.18% vs 12.21%). At $\alpha = 1.0$, it approaches unconditional geometry selection and becomes worse. Geometry is therefore a useful candidate signal, but a deployable gate needs richer confidence features.

To test whether this failure is merely due to a poor

hand-chosen threshold, we also train a tiny recency-aware gate. For each K , we fit a leave-one-session-out decision stump over geometry features such as distance ratio, absolute margin, source lag, and previous/source distances. This learned gate improves over previous-session recency only at $K = 20$, and only slightly (12.21% to 12.15% PER; Table 4). At $K = 5$ and $K = 10$, it overuses older sources and performs worse than previous. A one-dimensional learned gate is therefore still not enough; future gates need richer confidence signals.

The non-previous selections explain why the problem is difficult. Older sources beat the previous source in only a minority of non-previous geometry selections (3/11, 4/9, and 3/11 for $K = 5, 10, 20$ using phoneme-weighted session PER), and the Spearman correlation between geometry/previous distance ratio and older-source gain is weak ($\rho = 0.12, -0.02, 0.28$). Thus covariance geometry is useful for candidate retrieval, but not yet reliable as a standalone confidence measure.

To ask why older sources sometimes beat the previous

Table 5: Preliminary drift fingerprints for non-previous source choices. Negative principal-angle change means the older source is closer to the target subspace than the previous source.

K	Case	n	Δ principal angle
5	older wins	3	-1.37°
5	older loses	8	-0.06°
10	older wins	4	-1.93°
10	older loses	5	$+1.02^\circ$
20	older wins	3	-2.91°
20	older loses	8	$+0.55^\circ$

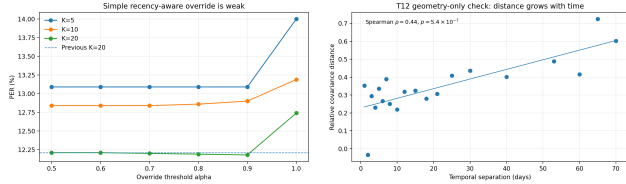


Figure 6: Left: a naive recency-aware geometry override does not substantially beat the previous-session baseline. Right: T12 diagnostic-block covariance distance increases with temporal separation, supporting the broader geometry framing outside T15.

session, we compare drift fingerprints for target–previous pairs and target–older-source pairs. This analysis is preliminary because older-source wins are rare, but it reveals a consistent pattern: wins are better explained by subspace geometry than by mean/scale shifts alone. When older sources win, they reduce the mean principal angle relative to the previous source by 1.37° , 1.93° , and 2.91° for $K = 5, 10, 20$. When older sources lose, this subspace improvement is absent or reverses (Table 5). This suggests that non-recent input states are useful when they better match the target neural subspace, supporting a drift-typing view of the adaptation ladder.

5.5 Geometry-Only Supporting Checks

The main claims above are decoder-facing T15 results. We use T12 and inner-speech data only as geometry-only supporting checks. In T12 diagnostic blocks, covariance distance increases with temporal separation for spike-power features (Spearman $\rho = 0.44$, $p = 5.4 \times 10^{-7}$), and threshold-crossing variants tx1–tx4 show the same positive trend ($\rho = 0.34$ – 0.47 , all $p < 1.3 \times 10^{-4}$). Geometry selects an older non-previous source in 6–9 of 15 targets depending on feature type, including a 53-day lag in one spike-power case. This supports the idea that recency is important but not sufficient to describe longitudinal neural geometry, without claiming T12 decoder replication.

As a separate behavior-state check, we analyzed interleaved verbal-behavior blocks from the T12/T15/T16/T17 inner-speech dataset [Kunz et al., 2025]. We did not train a decoder, evaluate PER, or at-

Table 6: Deployment interpretation of the adaptation ladder. The goal is not to apply every adaptation step every day, but to choose the lightest action that is supported by the beginning-of-day evidence.

Situation	Suggested action	Evidence in this audit
Recent previous	Reuse previous	strongest baseline
Fixed-source mismatch	Avoid fixed source	22.67–34.43% PER
Mean/scale shift only	Moment correction insufficient	$\sim 22.7\%$ PER
Labeled K trials	Calibrate input	22.67% $\rightarrow$ 18.59% PER
Non-recent match	Retrieve older state	lower subspace mismatch
Behavior shift	Index behavior	inner-speech geometry

tempt privacy-sensitive inner-speech decoding. Within each participant, we grouped activity by speech behavior and asked whether covariance geometry retrieves the matching behavior. The strongest setting, binned threshold crossings during the go epoch, reached 25/26 nearest-behavior accuracy (96.15% vs. 20% chance), with between-behavior distances 2.60 times within-behavior distances. This preliminary check suggests that input states may reflect behavior mode as well as recording day.

6 Discussion

The main lesson is that cross-session adaptation should not be treated as a single fixed correction. Wrong input states cause large decoder-facing degradation, moment matching does not emulate the learned day-specific input layers, and learned input calibration recovers only part of the lost performance. The most useful low-cost intervention is to choose a compatible input state while keeping the GRU decoder frozen.

The resulting deployment view is conservative. A new day should first try the previous input state, because it is the strongest simple default when it is truly recent. Geometry should then be used to retrieve older candidates only when short-window covariance/subspace structure suggests a better match. If no stored state is reliable and labels are available, residual input calibration can update only the input transform. Table 6 summarizes this interpretation: choose the lightest action supported by the beginning-of-day evidence.

Geometry is therefore best viewed as a retrieval signal, not as a standalone policy. The failed distance-ratio and stump gates show that covariance distance alone is too weak to safely override recency. The drift-fingerprint analysis points to a better direction: older states are useful when they reduce subspace mismatch. A future online policy should combine temporal gap, subspace margin, source rank, session quality, decoder confidence, and possibly hidden-state or logit-space diagnostics. Preliminary alignment probes should also be treated cautiously: geometry-based feature transformations may introduce harms, so geometry is currently more reliable for retrieval than for automatic feature transformation.

A small number of difficult sessions dominate worst-

case PER, especially 2025-03-30 and 2025-03-14. Sensitivity checks excluding the worst one or two sessions preserve the qualitative ordering, so we retain all sessions and treat these days as stress cases rather than exclusions. Finally, the geometry-only T12 and inner-speech checks suggest that the same state-library view may extend beyond T15 recording days to other participants and behavior modes, but decoder-facing replication, WER evaluation, and closed-loop adaptation remain future work.

7 Limitations

This study is a decoder-facing audit, not a new decoding architecture. Greedy PER is lighter than full WER and avoids the language-model/Redis evaluation stack. T12 is geometry-only and should not be interpreted as T12 PER replication. The inner-speech analysis is also geometry-only; it does not decode imagined speech or evaluate a communication system. The learned adapter and K-shot source-selection experiments simulate a beginning-of-day calibration window within the validation split and evaluate only on the remaining trials. They are intended to map adaptation rungs, not to optimize a final deployment benchmark. We focus the main claims on $K \leq 20$; larger calibration windows can be informative as feasibility checks, but in this validation split they consume a large fraction of the available session and are therefore not treated as minimal calibration.

8 Conclusion

For T15 brain-to-text decoding, fixed non-native input layers break the decoder, moment correction is too weak, and learned input adaptation improves PER only partially. The largest practical improvement comes from selecting a compatible past input layer. The immediately previous session is a strong default, while drift geometry is a measurable signal for deciding when that default may need to be overridden. This supports a recency-aware, geometry-guided adaptation ladder: start with the previous input state, measure drift geometry from a short unlabeled window, and move to stronger calibration only when the geometry indicates that simple source selection is not enough.

Ethical Statement

This work uses publicly released intracortical BCI datasets and reports only aggregate analyses. It does not attempt to infer private inner speech or deploy a decoder. Future extensions to inner-speech or cross-behavior state selection should include explicit intent-gating and privacy safeguards.

References

- [Card et al., 2024] Nicholas S. Card, Maitreyee Wairagkar, Carrina Iacobacci, et al. An accurate and rapidly calibrating speech neuroprosthesis. *New England Journal of Medicine*, 391(7):609–618, 2024.
- [Degenhart et al., 2020] Alan D. Degenhart, William E. Bishop, Emily R. Oby, et al. Stabilization of a brain-computer interface via the alignment of low-dimensional spaces of neural activity. *Nature Biomedical Engineering*, 4:672–685, 2020.
- [Engelhard et al., 2019] Ben Engelhard, Ran Darshan, Nofar Ozeri-Engelhard, Zvi Israel, Uri Werner-Reiss, David Hansel, Hagai Bergman, and Eilon Vaadia. Neuronal activity and learning in local cortical networks are modulated by the action-perception state. *bioRxiv*, 2019.
- [Fan et al., 2023] Chaofei Fan, Nick Hahn, Faheem Kamdar, et al. Plug-and-play stability for intracortical brain-computer interfaces: a one-year demonstration of seamless brain-to-text communication. In *Advances in Neural Information Processing Systems*, 2023.
- [Fiedler et al., 2025] Tobias Fiedler, Leon Hermann, Florian Müller, Sarel Cohen, Peter Chin, Tobias Friedrich, and Eilon Vaadia. Teaching Wav2Vec2 the language of the brain. *arXiv preprint arXiv:2501.09459*, 2025.
- [Karpowicz et al., 2025] Bryan M. Karpowicz, Yahia H. Ali, L. N. Wimalasena, et al. Stabilizing brain-computer interfaces through alignment of latent dynamics. *Nature Communications*, 16:4662, 2025.
- [Kunz et al., 2025] Erin M. Kunz, Benyamin A. Krasa, Faheem Kamdar, et al. Inner speech in motor cortex and implications for speech neuroprostheses. *Cell*, 188(17):4658–4673.e17, 2025.
- [Mandelblat-Cerf et al., 2011] Yael Mandelblat-Cerf, Itai Novick, Rony Paz, Yuval Link, Sharon Freeman, and Eilon Vaadia. The neuronal basis of long-term sensorimotor learning. *Journal of Neuroscience*, 31(1):300–313, 2011.
- [Vaadia and Birbaumer, 2009] Eilon Vaadia and Niels Birbaumer. Grand challenges of brain computer interfaces in the years to come. *Frontiers in Neuroscience*, 3:151–154, 2009.
- [Willett et al., 2021] Francis R. Willett, Donald T. Avansino, Leigh R. Hochberg, Jaimie M. Henderson, and Krishna V. Shenoy. High-performance brain-to-text communication via handwriting. *Nature*, 593:249–254, 2021.
- [Willett et al., 2023] Francis R. Willett, Erin M. Kunz, Chaofei Fan, et al. A high-performance speech neuroprosthesis. *Nature*, 620:1031–1036, 2023.
- [Zhang et al., 2026] Zhanqi Zhang, Shun Li, Bernardo L. Sabatini, Mikio Aoi, and Gal Mishne.

Align: Adversarial learning for generalizable speech neuroprosthesis. *arXiv preprint arXiv:2603.18299*, 2026.